MODULE: INTRODUCTION

Fundamental differences between Science & Engineering:

Science is the intellectual and practical activity encompassing the systematic study of the structure and behaviour of the physical and natural world through observation and experiment.
Engineering is the branch of science and technology concerned with the design, building and use of engines, machines and structures.

Science	Engineering
1. It is synthesis of knowledge by understanding nature's law.	1. It is application of knowledge to transform nature for serving people.
2. Scientists have a sound knowledge of science, mathematics and technology.	2. Engineers are trained to use these principles in designing creative solutions to the challenges.
3. Science studies what is.	3. Engineering creates what never was.
4. Science is about knowing.	4. Engineering is about doing.

Comparison between human eye and camera:

Human eye	Camera
1. Amount of light that get through to the lens is controlled by the pupil.	1. Amount of light that get through to the lens is controlled by the shutter.
2. Lens system is a flexible bi-convex lens.	2. Lens system is a rigid bi-convex lens.
3. Image sensing is done on the retina.	3. Image sensing is done on the film or sensor.
4. Here, image distance (distance between lens and retina) is fixed. So, if the object distance is changed, then the focal length is changed to create sharp image. This is done by changing the shape of the lens; a muscle called ciliary muscle does the job.	4. Here, lens has fixed focal length. If the object distance is changed, the image distance is adjusted by moving the lens.

Comparison between bird flying and aircraft:

Bird flying	Aircraft
1. Birds flap their wings to fly.	1. Airplane have engine, wiring system, cables and fuel to fly.
2. Birds use metabolic energy (ATP) to fly.	2. Airplane use fuel energy to fly.
3. Bird's wings provide lift and thrust.	3. Airplane's wings have an airfoil shape that only provides lift (following Bernoulli principle).
4. Birds can change their wing span's shape.	4. Airplane can twist and turn but cannot change their wing span's shape.

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Most exciting aspect of biology as an independent scientific discipline:

Biology is the branch of science which deals with the study of the various aspects of living beings. It is a science that seeks to understand how living things perform their life activities. Therefore, it is more appropriately called 'Life Science'.
Though the definition of biology appears very simple, the branch is equally complex and extensive, as it deals with the various exciting aspects of the enormous number of living organisms spread in time and space. The branch deals with –

- Structural organisation of living organisms.
- Functions of their body parts.
- Habitats where they live.
- Their relationship and interaction among themselves as well as surrounding environment.
- How they pass their specific characters to the next generations.
- How they have evolved from primitive to more advance forms.

Why we need to study biology?

Following are some of the many reasons to study biology –

- Biology has made us aware of the structure and function of human's own body, organs and systems.
- Biology has provided improved food, comforts and luxury by acquiring knowledge about beneficial plants and animals.
- Biology has led to the discovery and development of new and better varieties of crop, to solve food problem.
- Biology has helped in discovering causes and methods of eradication of disease causing germs.
- Biology has helped us to discover and manufacture medicine for treating various diseases.
- Biology teaches us to conserve natural resources, forests, wild animals, etc. for maintaining environmental equilibrium.
- Biology has helped us to understand intricate relationship that exists among organisms in an ecosystem.
- Biology reminds us our role and responsibility to protect our physical and biological environment.
- Biology has made us aware of the environmental hazards caused by our activities like discharge of pollutants, unmindful wastage of natural resources, indiscriminate felling of trees, etc.
- Biology enhances our faculty of appreciation of the beauty of rich biodiversity.

Biological observations of 18th century that led to major discoveries (example from Brownian motion and origin of Thermodynamics, by referring to the original observation of Robert Brown and Julius Mayer):

Discovery of Brownian motion and original observation of Robert Brown:

Brownian motion (random motion of particles suspended in a fluid resulting from their collison with fast moving molecules in the fluid) is named after the botanist Robert Brown,

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who first discovered this phenomenon in 1827, while looking through a microscope at pollen of the plant *Clarkia pulchella* immersed in water. In 1905, almost 80 years later, physicist Albert Einstein published a research paper, where he modelled the motion of pollen as being moved by individual water molecules, making one of his first big contribution to science. This explanation of Brownian motion served as convincing evidence that atoms and molecules exist and was further verified experimentally by Jean Perrin in 1908. Perrin was awarded Nobel Prize in Physics in 1926.

Observation of Julius Robert von Mayer and origin of Thermodynamics:

Julius R. Mayer was a German physician. He is best known for enunciating in 1841, one of the original statements of the conservation of energy, or what is known as one of the first version of 'First Law of Thermodynamics', namely, *energy can neither be created nor destroyed*. He proposed that plants convert light energy into chemical energy. In, 1842, Mayer described the vital chemical process, now referred to as oxidation as the primary source of energy for any living creature.

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